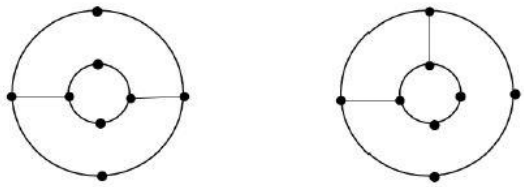
MAT 206 GRAPH THEORY

QUESTION BANK

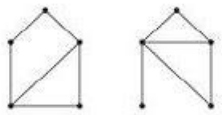
UNIT WISE QUESTION BANK

1. Prove that in a connected graph every $u - v$ walk contains a $u - v$ path.
2. Prepare a short note on walk, trial, path, and cycle of a connected graph.
3. Draw all simple graph of one, two, three, and for vertices.
4. Draw the graph of the following chemical compounds CH_4, C_2H_6, C_6H_6

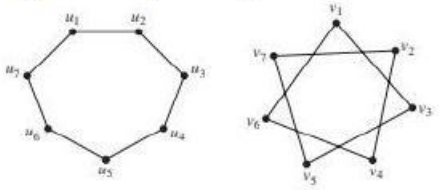
5. Show that the maximum number of edges in a simple graph with 'n' vertices is $n(n-1)/2$.
6. Prove that the maximum number of edges in a simple graph with 'n' vertices is $(n-1)$.
7. Draw a connected graph that becomes disconnected when any edge is removed from it.
8. Prove that a simple graph with "n" vertices must be connected if it has more than $(n-1)(n-2)/2$ edges.
9. Are the two graphs in the figure isomorphic.



10. Prove that any two simple connected graphs with 'n' vertices all of degree 2 are isomorphic.
11. What is the maximum number of edges in a simple graph with 'n' vertices?
12. There are 25 telephones in Metropolis. Is it possible to connect them with wires so that each telephone is connected with exactly 7 others? Why?
13. Define complete graph and complete bipartite graph. Draw a graph which is a complete graph as well as a bipartite graph.
14. Explain walk, path, and circuit with the help of examples.
15. Define isolated vertex, pendent vertex, even vertex and odd vertex. Draw a graph that contains all the above.
16. What is the number of vertices in an undirected connected graph with 27 edges,6 vertices of degree 2,3 vertices of degree 4, and the remaining of degree3?
17. Determine whether the graphs are isomorphic. If Yes or No justify



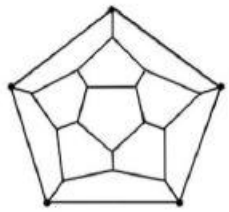
18. Determine whether the given pairs of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.



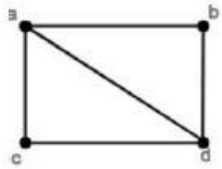
19. Explain seating problem in graph theory using graphical representation.
20. Can five houses be connected to two utilities without connections crossing? Justify your answer by drawing a graph.

UNIT WISE QUESTION BANK

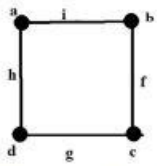
1. Is Hamiltonian circuit always implying Hamiltonian path? Is the converse true? Justify your answer.
2. Is the converse of Dirac's theorem true? Justify your answer.
3. Prove that if a connected graph G is decomposed into two sub graphs g_1 and g_2 there must be at least one vertex common between g_1 and g_2.
4. Is Peterson graph Hamiltonian. Justify your answer.
5. Does the following graph have a Hamiltonian Circuit?

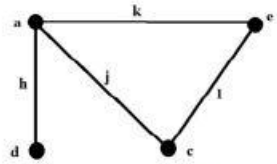


6. Check whether the graph is Eulerian graph. Justify your answer.



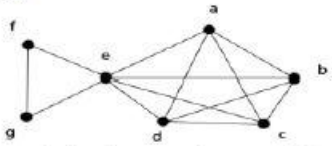
7. Find the union, intersection and ring sum of the following graphs.



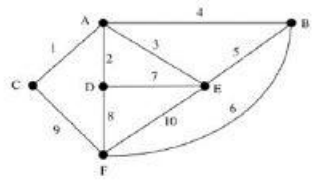


8. State travelling salesman problem. How it is related to Hamiltonian circuits?
9. For which values of m, n is the complete graph k_{mn} an Euler graph ? Justify your answer.
10. State Dirac's theorem and check the applicability of the following graph.

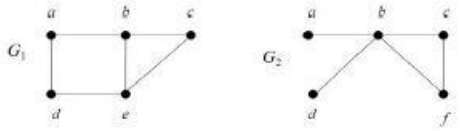
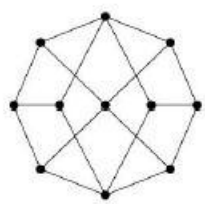
11. Define an Euler graph. Check whether the given graph is Euler graph and if yes, give the Euler walk. Justify your answer.



12. Find the smallest integer n such that the complete graph has at least 500 edges.
13. Draw a connected graph that becomes disconnected when any edge is removed is removed from it.
14. Define a Hamilton Cycle. Illustrate. Describe the Travelling Sales man Problem.
15. Fuse the vertices A and B of the following graph



16. Find the union, intersection, and ring sum of the following two graphs



17. For what value of 'n' do the complete graph K_n have an Euler circuit? Give the reason.
18. Consider the following graph, find a Hamiltonian path by labelling the vertices of the given graph, can your path be extended to a Hamiltonian circuit?